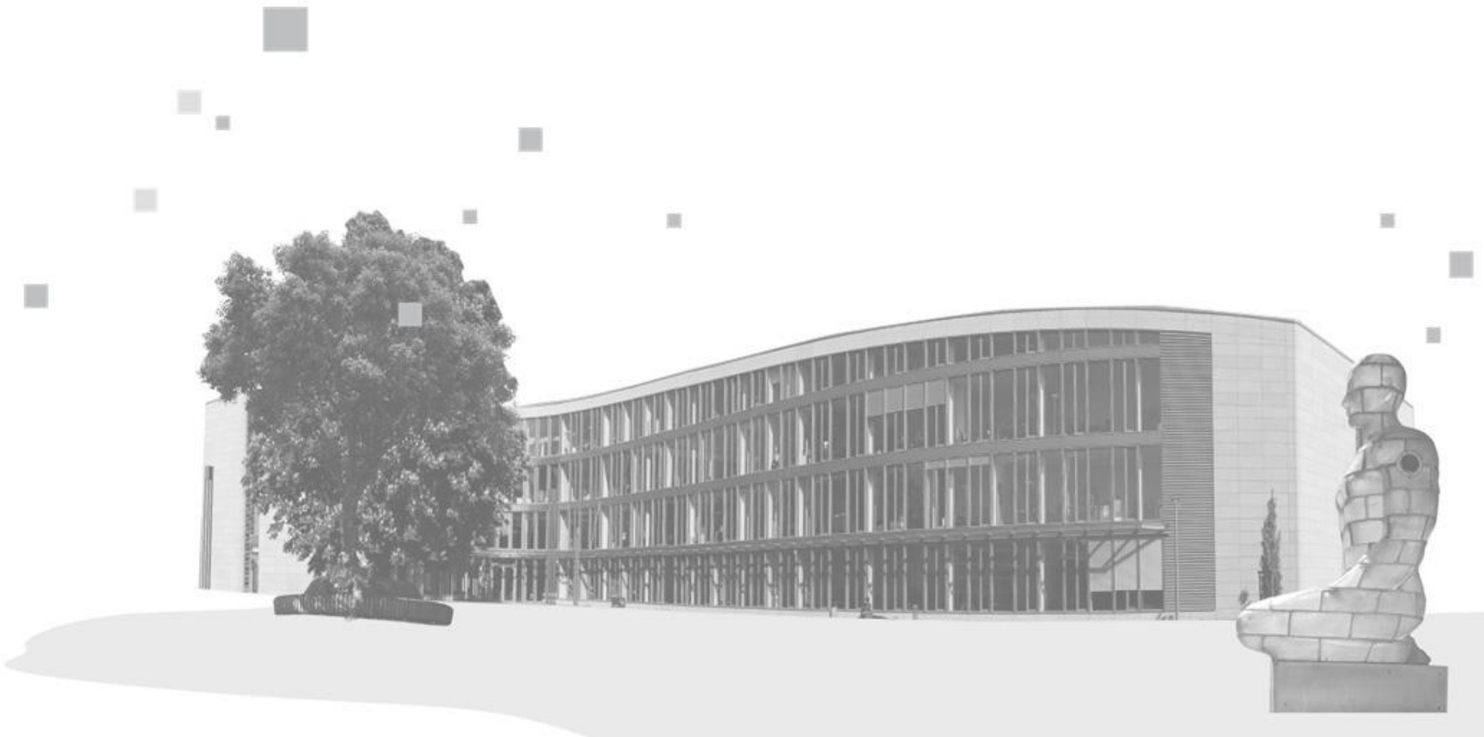
KI Service Zentrum by Hasso-Plattner-Institut

HPI

Reinforcement Learning I – How Agents Learn by Trial and Error

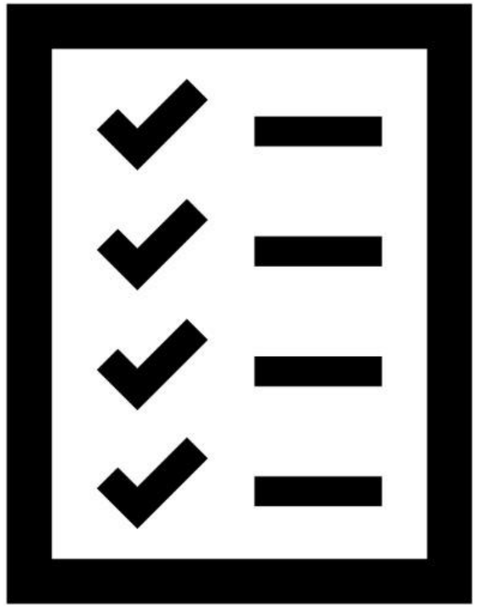
David Goll
Jill Barvencik



Design IT.
Create Knowledge.

www.hpi.de

Attendance List



Four AI Service Centres in Germany

KI Service Zentrum by Hasso-Plattner-Institut

https://hpi.de/ki-servicezentrum/

Overview

KI-Servicezentren

Goal:
Reduce barriers to the implementation of AI applications in society and the economy

Gefördert durch:

Bundesministerium für Forschung, Technologie und Raumfahrt

AI Service Centre Berlin-Brandenburg

KI Service Zentrum by Hasso-Plattner-Institut HPI

Structure

Gefördert durch:

Bundesministerium für Forschung, Technologie und Raumfahrt

Newsletter

Education

Talks

tele-task.de/series/1463

- Guest talks about research and innovation

Work shops

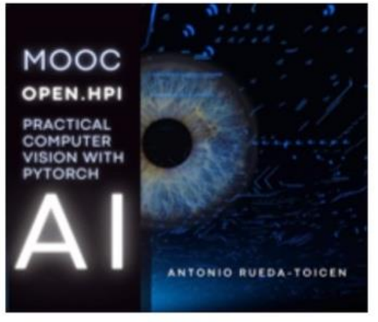
aimaker.community

- Practical topics
- Example topics: Speech2summary, Docker for ML, semantic search

MOOCs

open.hpi.de/channels/ai-service-center

- ChatGPT: What does generative AI mean for our society?
- Profitable AI
- Understanding and avoiding AI biases



Gefördert durch:
Bundesministerium für Forschung, Technologie und Raumfahrt

Sprechstunde buchen

Consulting

AI consultation hours

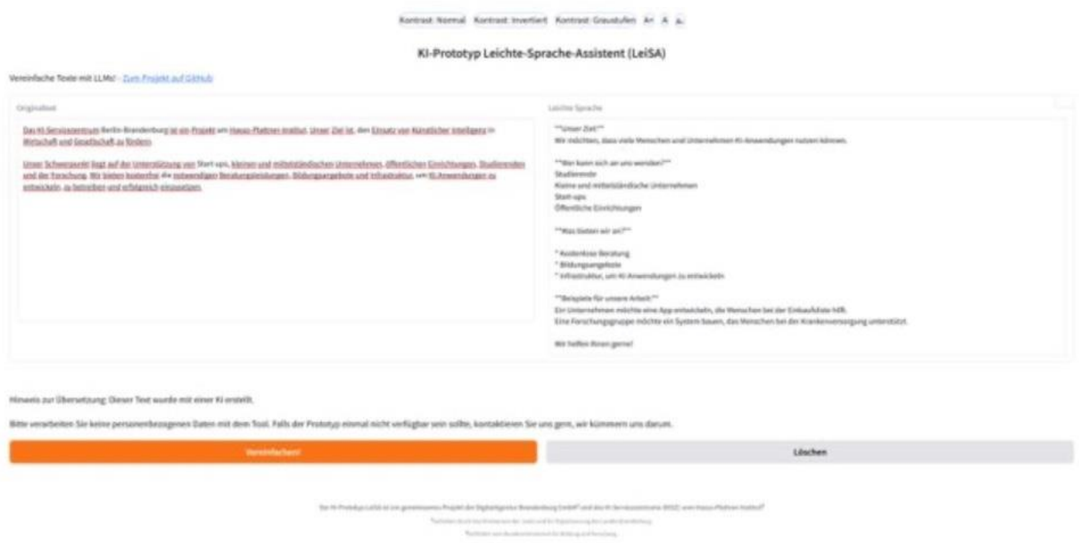
- Answering questions:
  - about AI infrastructure
  - about AI models & frameworks
  - about AI use cases

AI pilot projects

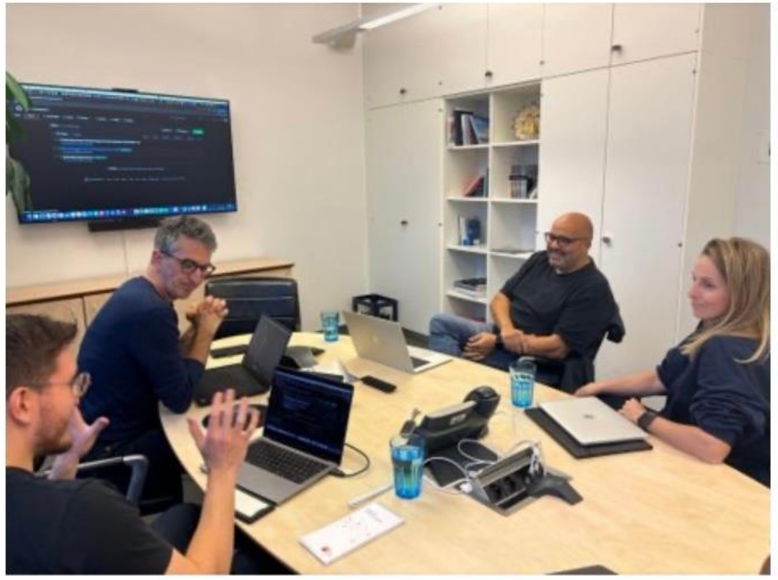
- Co-development of a prototype
- Application every three months
- Selection criteria, e.g. AI maturity, public benefit
- Publication of results

Cooperation

- Jointly organized network meetings



github.com/aihpi/leichte-sprache



Previous AI pilot projects

- Generation of math problems
- Generation of easy language
- Generation of upcycling suggestions
- Reduction of food waste
- Dating by handwriting

Jetzt bewerben!

Gefördert durch:
Bundesministerium für Forschung, Technologie und Raumfahrt

Zugangsanfrage
aisc.hpi.de

Infrastructure

- **Free access**
- No production operation
  - Data should be **anonymised** or **synthesised**
  - No **hosting** of products
- **Reporting** & **publication** by users
- **Old rights** remain with users
- **New rights** remain with users
  - Granting of rights of use for research and teaching

Training
- 64 NVIDIA H100 GPU

Inference
- 40 NVIDIA A30 GPU

ARM Server
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge
- ARMv8 CPU
- NVIDIA Jetson AGX Module

Neuromorph
- 288 SpiNNaker2 Chips

Storage
- 1.5 PB NVMe

Network
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

Gefördert durch:
Bundesministerium für Forschung, Technologie und Raumfahrt

Personal Introduction – David Goll

kisz@hpi.de
hpi.de/kisz

Background

Physics

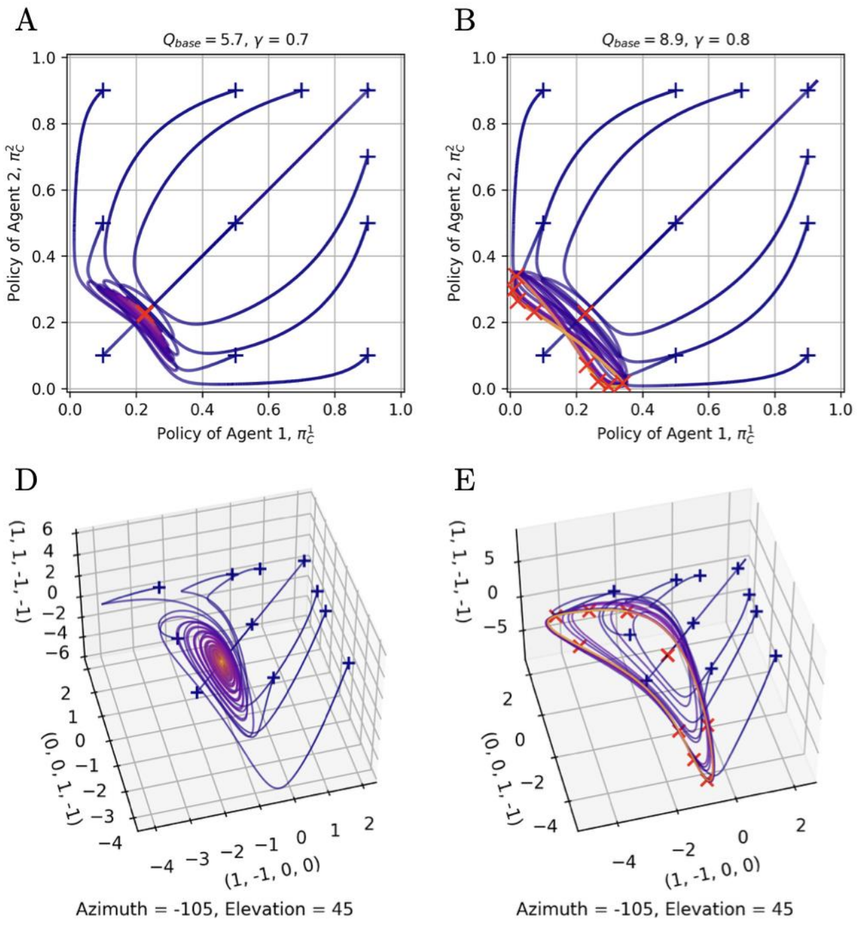
Master's thesis

"Analysis of Multi-Agent Reinforcement Learning from a Statistical Physics Perspective"

https://doi.org/10.3390/app16073524

Main message

Complex dynamics can arise even in simple MARL systems



Personal Introduction – Jill Barvencik

kisz@hpi.de
hpi.de/kisz

Background

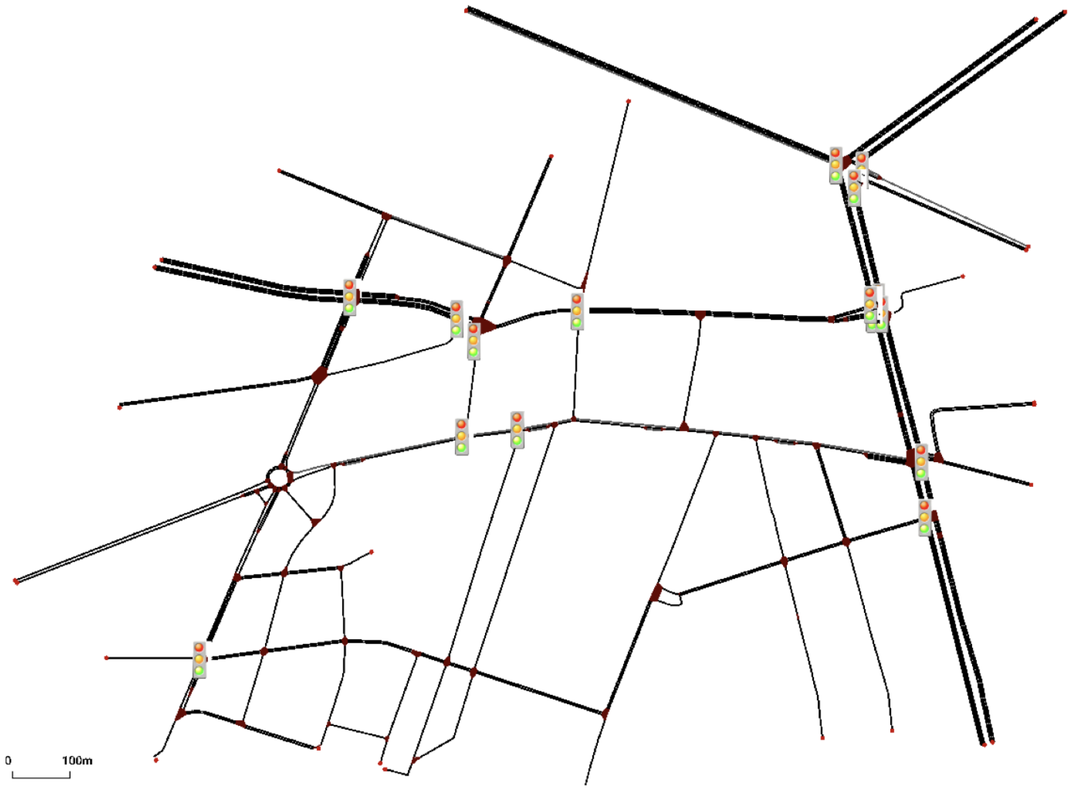
Mechanical Engineering and Computational Engineering Science

Master's thesis

"Deep Multi-Agent Reinforcement Learning for Real-World Large-Scale Traffic"

Main message

Used RL to stabilise seq2seq training and improve robustness beyond sparse reward signals



Personal Introduction – Your Turn

Please introduce yourself in 1–2 sentences

kisz@hpi.de
hpi.de/kisz

Two questions to get us started:

What is your background?

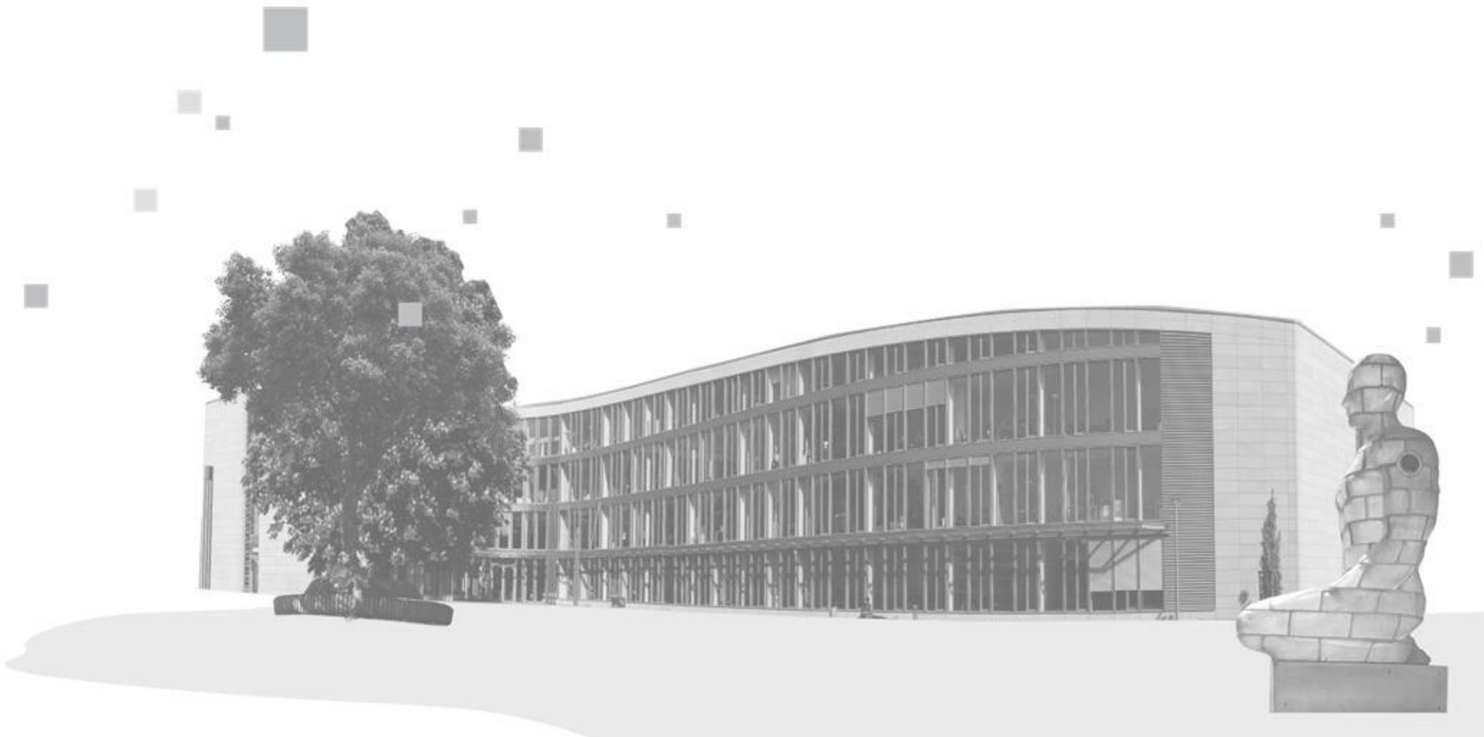
Why are you interested in RL?

KI Service Zentrum by Hasso-Plattner-Institut

HPI

Reinforcement Learning I – How Agents Learn by Trial and Error

David Goll
Jill Barvencik



Design IT.
Create Knowledge.

www.hpi.de

Agenda

What we will cover today

kisz@hpi.de
hpi.de/kisz

1. Introduction — core concepts, historical overview
2. Demonstrator — RL Lab

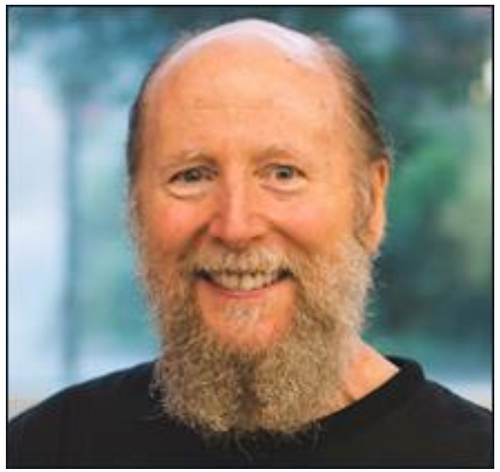
Break

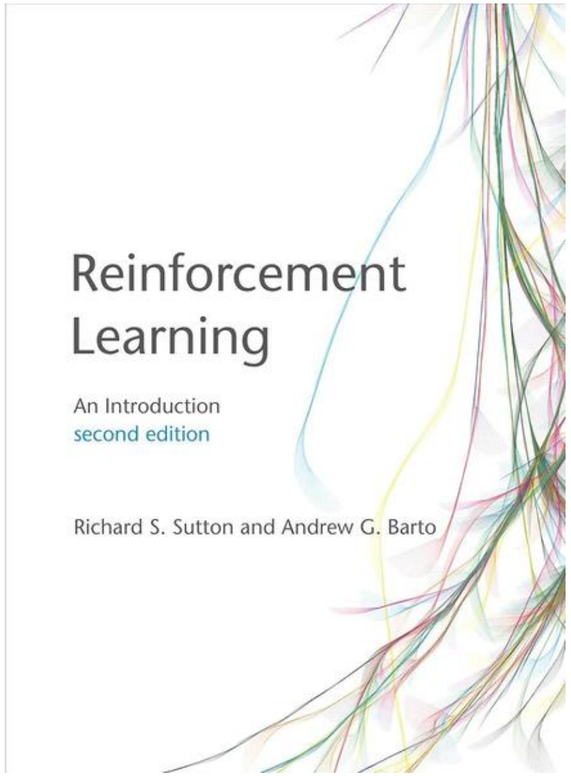
3. Interactive — Installation & Exploration of RL Lab
4. Today and Beyond — Modern Applications of RL
5. Interactive — Pen and Paper Exercise

Motivation and Context I

“Of all the forms of machine learning, reinforcement learning is the closest to the kind of learning that humans and other animals do.” – Sutton & Barto

kisz@hpi.de
hpi.de/kisz





Source of pictures: https://amturing.acm.org

Motivation and Context II

kisz@hpi.de
hpi.de/kisz

“Of all the forms of machine learning, reinforcement learning is the closest to the kind of learning that humans and other animals do.” – R. S. Sutton

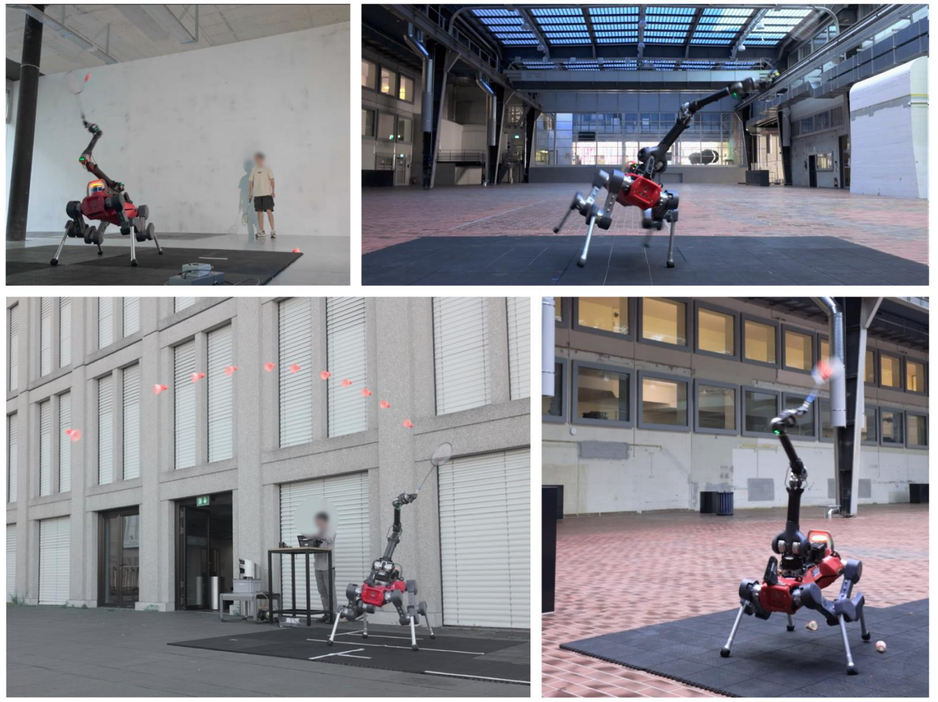


Fig. 1. Deployment of the badminton control policy on our legged manipulator. Our system operates entirely on onboard perception and computation for shuttlecock detection, trajectory prediction, and limb control. It has been tested in various environments, including the lab, a historic machine hall, and outdoor settings.

Video: ETH Zürich, https://www.youtube.com/watch?v=2hq-rWSowdo

Original paper: Ma et al., April 2025, https://arxiv.org/pdf/2505.22974v2

Motivation and Context III

Where reinforcement learning sits among the ML paradigms

kisz@hpi.de
hpi.de/kisz

Supervised Learning

- Models are **trained on labeled data** to predict outputs

Reinforcement Learning

- Learning optimal behavior by **interacting with an environment**

Unsupervised Learning

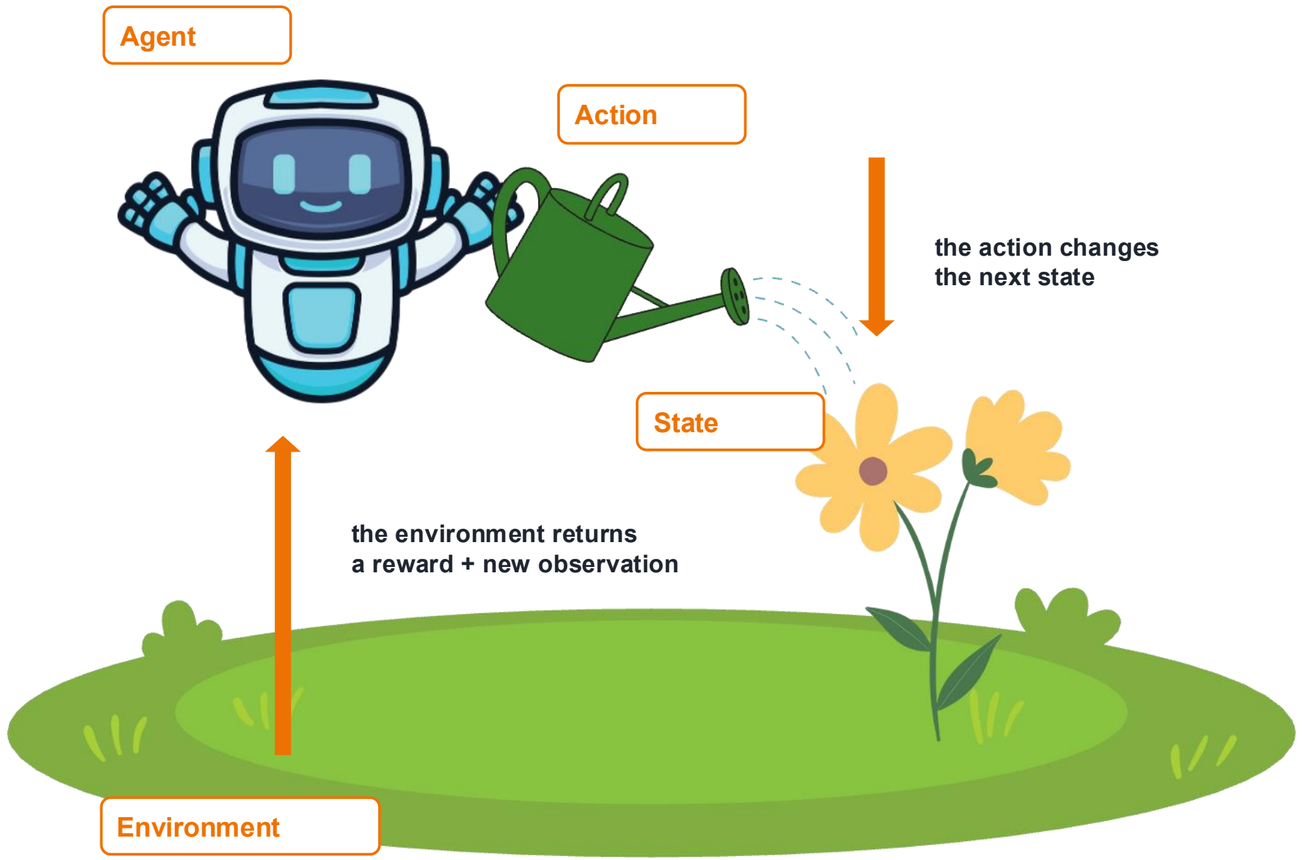
- Learn to **identify previously unknown patterns** and relationships in unlabeled data

used when **optimal actions are too complex or unknown to define** but can be learned through experience

Key RL Terms

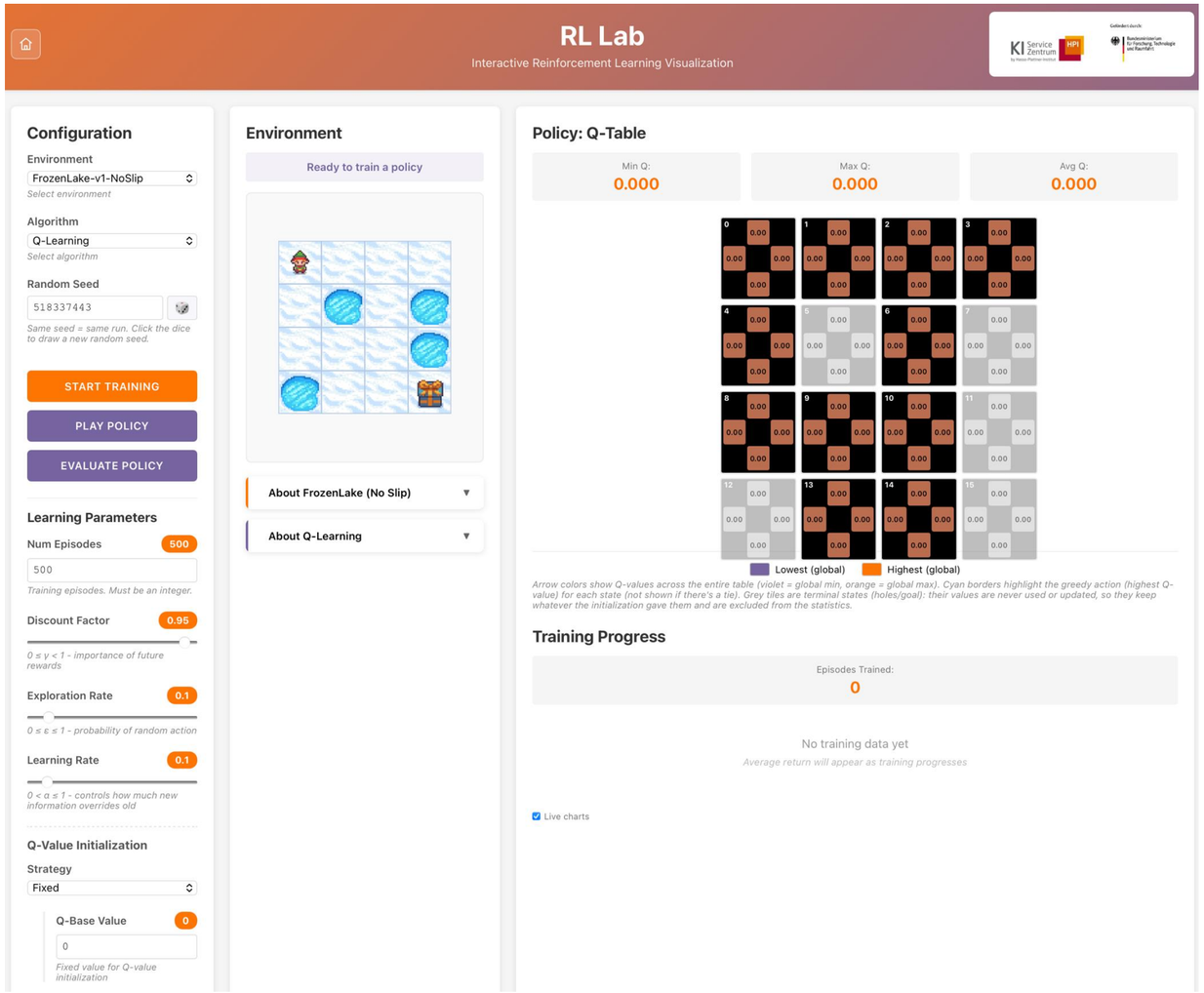
How an agent learns by interacting with its world

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz



- **Agent**
  the decision maker
- **Environment**
  the world it interacts with
- **State**
  the current situation
- **Action**
  what the agent can do
- **Reward**
  feedback from the environment
- **Policy**
  how the agent chooses actions

RL Lab demo: Untrained Agent



How Did We Get Here?

Four questions that moved reinforcement learning forward

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

1950s–70s

Foundations

How do we optimize decisions over time?

1980s–90s

Learning from Experience

What if the model is unknown?

1990s–2015

Deep RL

How does RL scale to complex states?

2016–2026

Today & Beyond

Where is RL today?

1950s–70s
Foundations
MDP, Bellman, DP

1980s–90s
Learning from Experience
TD and Q Learning

1990s–2015
Deep RL
Networks estimate values

2016–2026
Today & Beyond
RL in the real world

1950s–70s

How Do We Optimize Decisions Over Time?

MDP · Bellman · Dynamic Programming

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

MDP: a mathematical description of the world as a decision problem

s	current state the agent is in
s'	next state it moves to
a	action the agent takes
R(s,a)	reward it gets for that action

Bellman equation: links the value of today to reward now and future value

value of state s

=

reward right now

+

value of state s'

Dynamic Programming: computes the best action by sweeping all states, if the whole world is known

every state, action and reward known

↓

value of every state

↓

best action

1950s–70s
Foundations
MDP, Bellman, DP

1980s–90s
Learning from Experience
TD and Q Learning

1990s–2015
Deep RL
Networks estimate values

2016–2026
Today & Beyond
RL in the real world

1950s–70s

How Do We Optimize Decisions Over Time?

MDP · Bellman · Dynamic Programming

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

MDP: a mathematical description of the world as a decision problem

- **s** — current state the agent is in
- **a** — action the agent takes
- **R(s,a)** — reward it gets for that action
- **s'** — next state it moves to

Bellman equation: links the value of today to reward now and future value

value of state s

=

reward right now

+

value of state s'

Dynamic Programming: computes the best action by sweeping all states, if the whole world is known

every state, action and reward known

↓

value of every state

↓

best action

1950s–70s
Foundations
MDP, Bellman, DP

1980s–90s
Learning from Experience
TD and Q Learning

1990s–2015
Deep RL
Networks estimate values

2016–2026
Today & Beyond
RL in the real world

1950s–70s

How Do We Optimize Decisions Over Time?

MDP · Bellman · Dynamic Programming

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

MDP: a mathematical description of the world as a decision problem

- **s** — current state the agent is in
- **a** — action the agent takes
- **R(s,a)** — reward it gets for that action
- **s'** — next state it moves to

Bellman equation: links the value of today to reward now and future value

value of state s

=

reward right now

+

value of state s'

Dynamic Programming: computes the best action by sweeping all states, if the whole world is known

Know what can happen next
transitions + rewards

↓

Use Bellman to calculate how valuable each state is

↓

Choose the action with the highest expected value

How Did We Get Here?

Four questions that moved reinforcement learning forward

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

1950s–70s

Foundations

How do we optimize decisions over time?

1980s–90s

Learning from Experience

What if the model is unknown?

1990s–2015

Deep RL

How does RL scale to complex states?

2016–2026

Today & Beyond

Where is RL today?

1950s–70s
Foundations
MDP, Bellman, DP

1980s–90s
Learning from Experience
TD and Q Learning

1990s–2015
Deep RL
Networks estimate values

2016–2026
Today & Beyond
RL in the real world

1980s–90s

Can We Learn Without Knowing the Model?

TD Learning · Q-Learning

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

both learn from experienced rewards and transitions, and update their estimates step by step

TD Learning

learns how good a state is

V(s): the value of a state

↓

bootstrapping: updates an estimate using the next estimate

↓

improved estimate of V(s) after gaining enough experience

→ **Q-Learning (1992)**

Q-Learning Formula

how a Q-value is updated from experience

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

$$Q'(s,a) = Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

new Q-value	current Q-value	new information

Formula symbols

Q′(s,a)	updated value	**r**	reward	**s**	state
Q(s,a)	current value estimate	**γ**	discount factor	**a**	action
α	learning rate	**max Q(s′,a′)**	best next-state value	**s′**	next state

What learning parameters are there?

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

$$Q'(s,a) = Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

$$a = \begin{cases} \arg\max_{a'} Q(s,a') & \text{with probability } 1-\varepsilon, \\ \text{random action} & \text{with probability } \varepsilon. \end{cases}$$

α learning rate

determines how strongly new information influences the previous Q-value

γ discount factor

weights future rewards relative to immediate rewards

ε exploration rate

indicates how often random actions are chosen instead of the current best action

number of episodes

number of learning cycles the model goes through

initialization of Q-values

initial values of the Q-table

Example: Frozen Lake

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

Q-table initialized at 0, $\alpha = 0.1$, $\gamma = 0.95$, $r = +1$ at the goal, else 0

$$Q'(s,a) = Q(s,a) + \alpha \left[\underbrace{r + \gamma \max_{a'} Q(s',a')}_{\text{just } r \text{ if the episode ended}} - Q(s,a) \right]$$

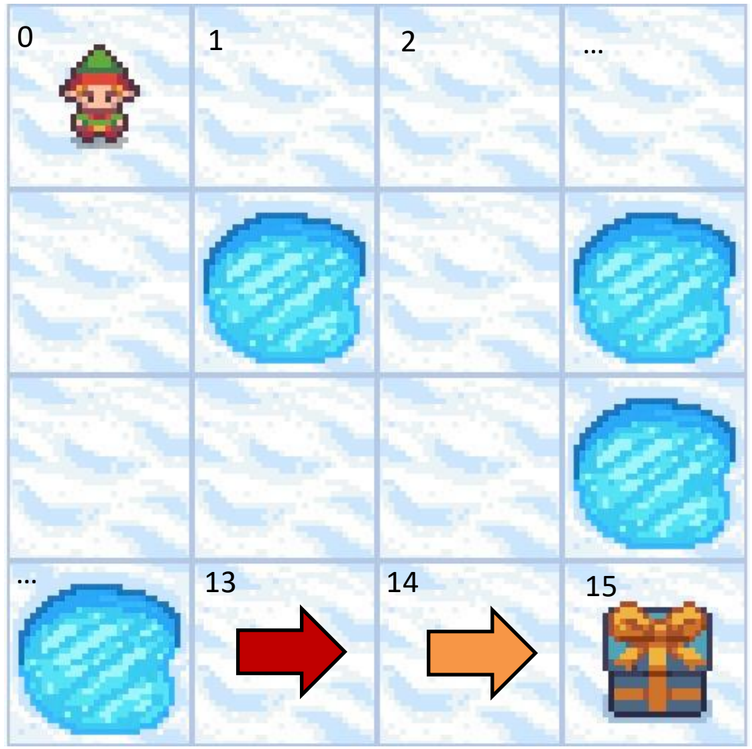
UPDATE 1 – reaching the goal by moving to the right in state 14

$$Q'(14, \rightarrow) = 0 + 0.1 * (1 - 0) = 0.1$$

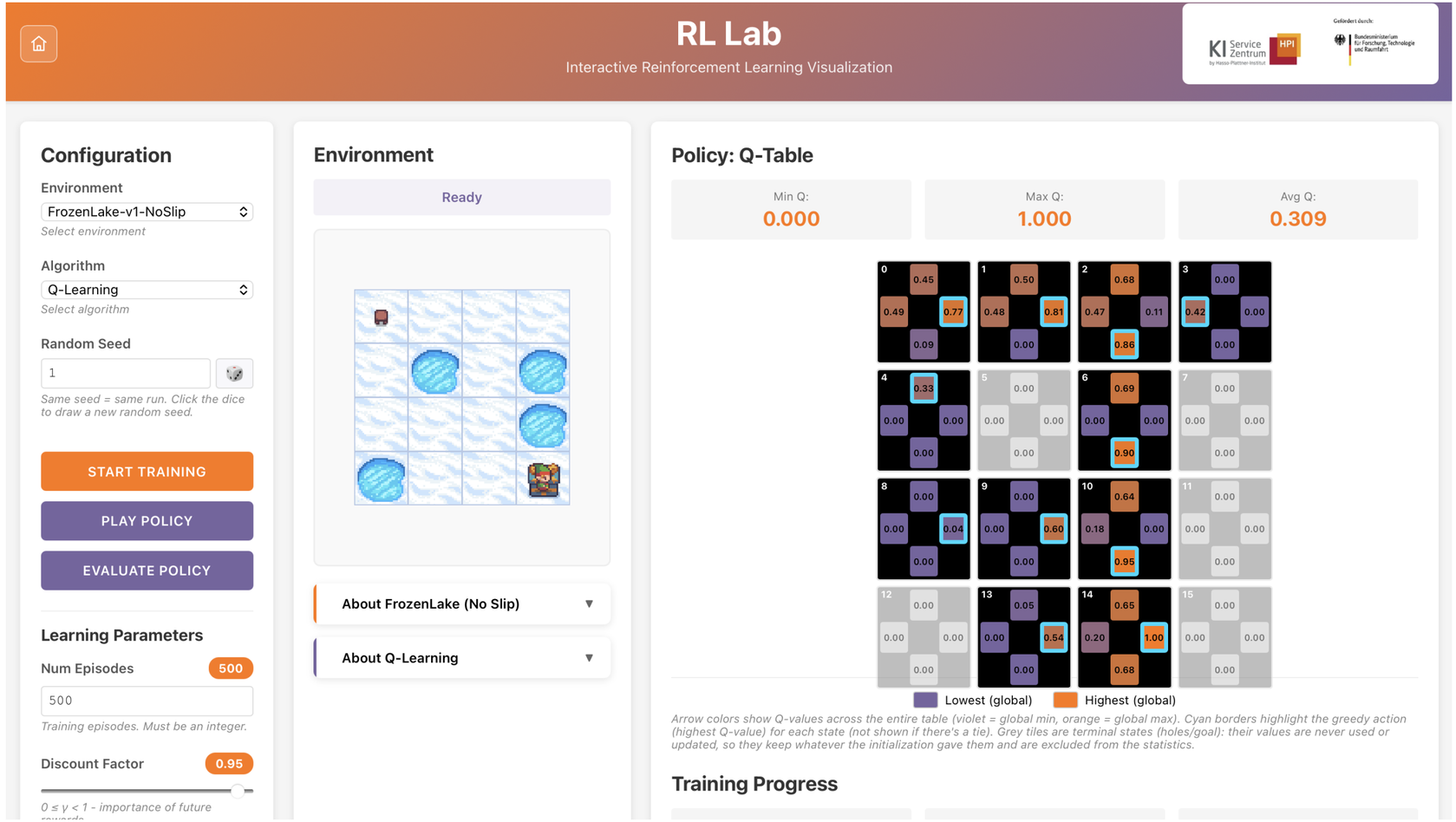
UPDATE 2 – in the next episode, when visiting state 13 and moving to the right into state 14

$$Q'(13, \rightarrow) = 0 + 0.1 * (0 + 0.95 * 0.1 - 0) = 0.0095$$

…



RL Lab demo: Q-learning Training



Install RL Lab

Please follow the instructions on

https://github.com/aihpi/workshop-rl1-introduction

Quick start:

```
git clone https://github.com/aihpi/workshop-rl1-introduction.git
cd workshop-rl1-introduction
docker compose pull
docker compose up -d
```

Then open **`http://localhost:3030`**

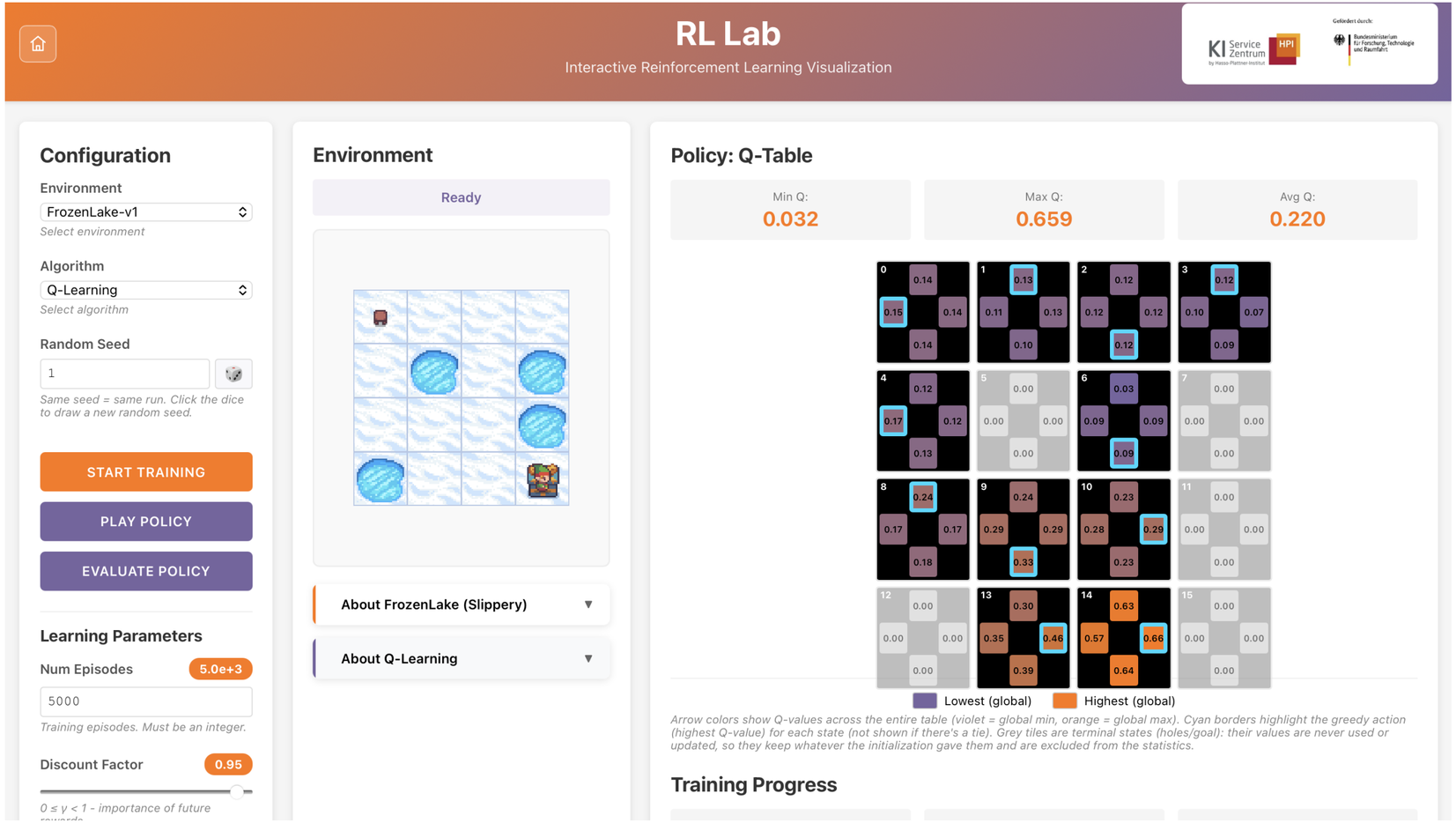
Ask us as soon as something fails, we try to get everyone running before the break.

KI Service Zentrum
by Hasso-Plattner-Institut
HPI

kisz@hpi.de
hpi.de/kisz

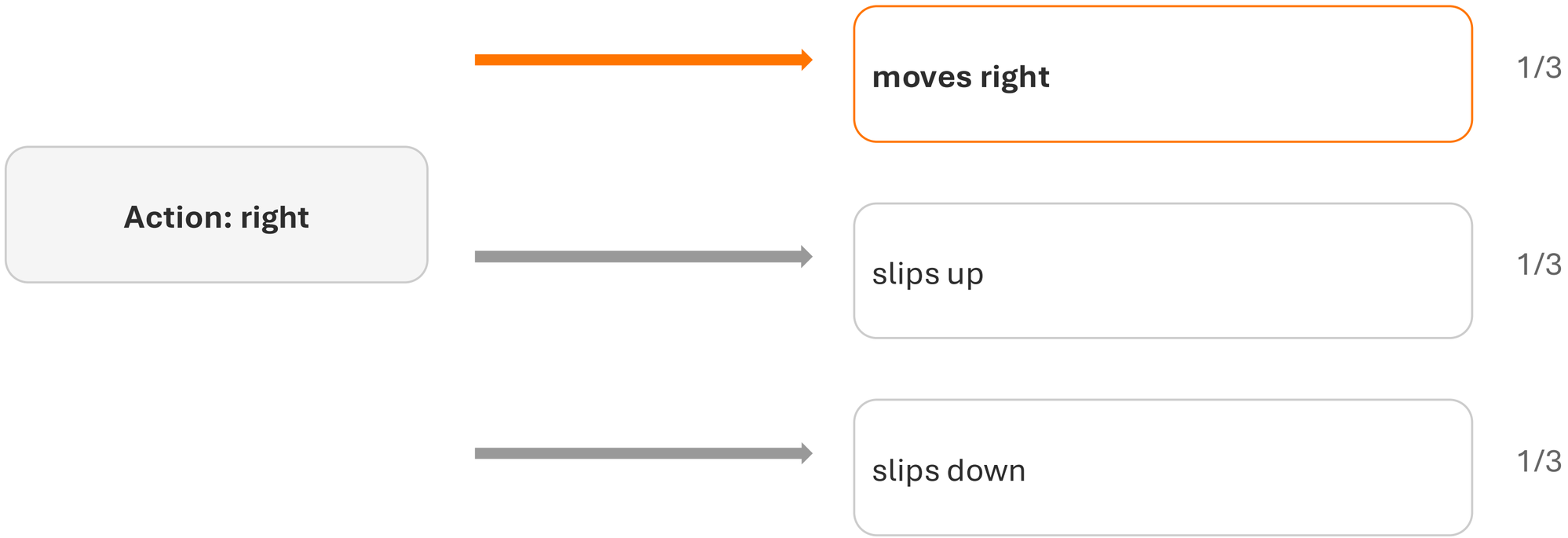
Break

RL Lab demo: Slippery FrozenLake



When the ice is slippery

The same action no longer leads to the same next state.



Learning takes longer, and the best policy can look counterintuitive.

How Did We Get Here?

Four questions that moved reinforcement learning forward

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

1950s–70s

Foundations

How do we optimize decisions over time?

1980s–90s

Learning from Experience

What if the model is unknown?

1990s–2015

Deep RL

How does RL scale to complex states?

2016–2026

Today & Beyond

Where is RL today?

1950s–70s
Foundations
MDP, Bellman, DP

1980s–90s
Learning from Experience
TD and Q Learning

1990s–2015
Deep RL
Networks estimate values

2016–2026
Today & Beyond
RL in the real world

1990s–2015

How Does RL Scale to Complex States?

Deep RL · DQN · Function Approximation

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

Problem

a Q-table only works for small, discrete state spaces

fine for a handful of discrete states

↓

Continuous state/action space

↓

the table no longer fits

DQN

a deep neural network approximates the Q-function

the network predicts Q(s,a) for any state

↓

the RL idea stays exactly the same

↓

the net replaces the table as function approximator

From Q-Learning to DQN

Same idea, same update — a network estimates Q(s, a)

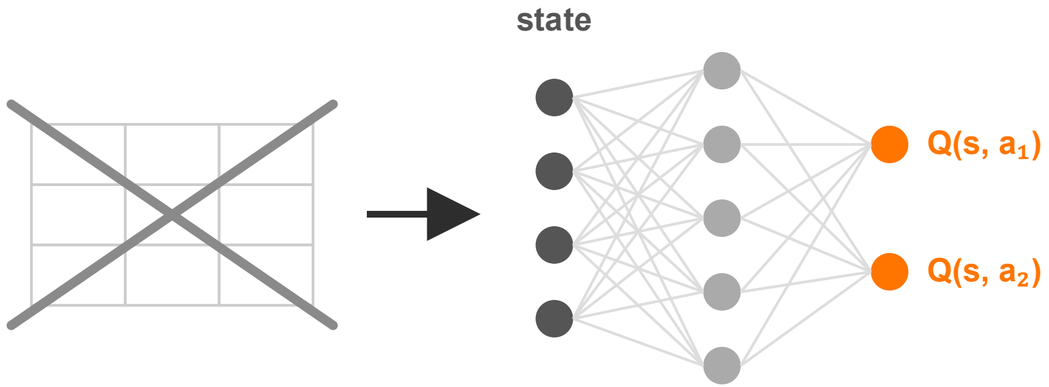
KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

CartPole: 4 continuous numbers

→ infinitely many states

→ **no table can hold them**



TWO TRICKS MAKE IT STABLE

1 Replay Buffer

a memory of experiences (s, a, r, s') — train on random batches

→ one lucky success teaches the network thousands of times

2 Target Network

a frozen copy computes the targets $r + \gamma \cdot \max Q_{target}(s', a')$

→ no chasing your own moving estimates

Same update rule, as a **gradient step**: loss = (target − Q(s,a))²

backpropagation carries the error backwards through the layers — every weight is nudged in proportion to its share of the error

Shared weights: every update moves ALL states a little

DQN = Q-Learning + network + replay buffer + target network

DQN on FrozenLake

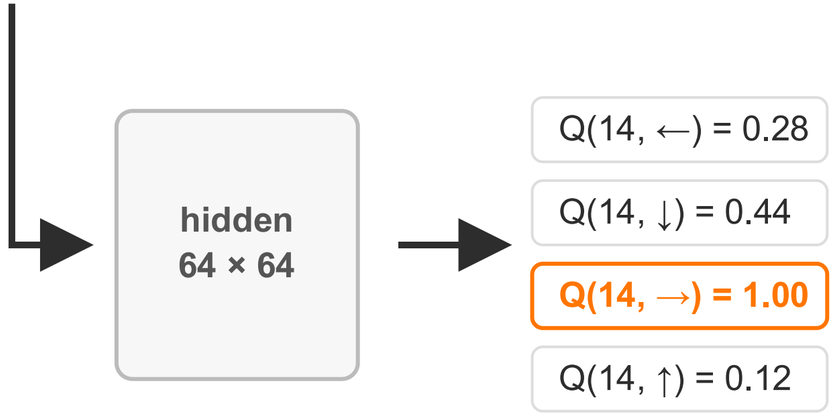
Only 16 states — ask the network all 16 questions, draw the answers as a table

KI Service Zentrum by Hasso-Plattner-Institut HPI

kisz@hpi.de
hpi.de/kisz

state 14 as a **one-hot vector**:

index 14



All 16 states → **the familiar Q-table** (what RL Lab shows)

Q-Learning's table

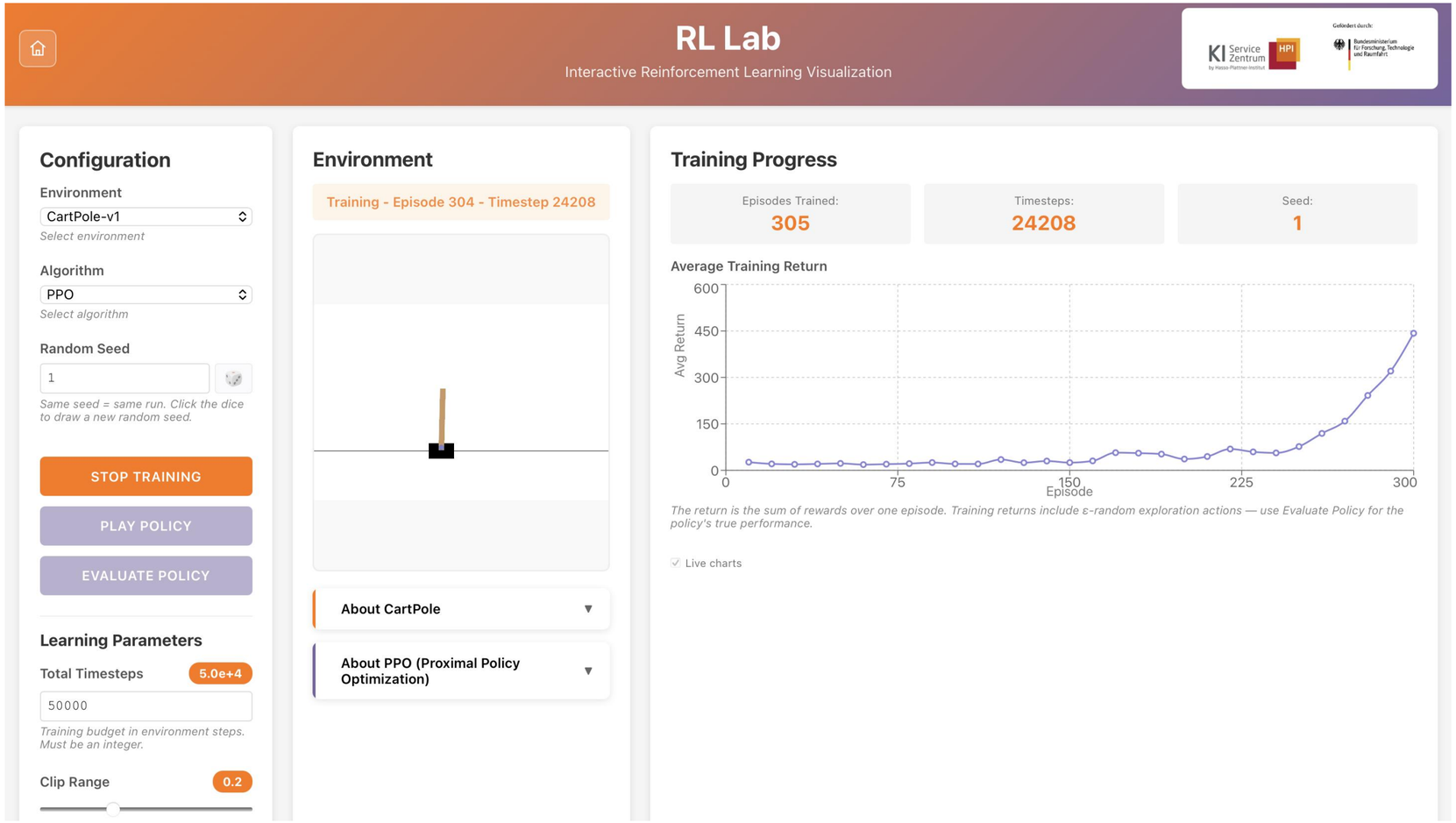
- every cell independent
- one update touches one cell
- values spread cell by cell

DQN's network

- all values share the weights
- every update moves ALL values
- terminal states: never trained, never used

RL Lab demo:

- DQN (alg)
- CartPole (env)
- PPO (alg)
- MountainCar (env)



How Did We Get Here?

Four questions that moved reinforcement learning forward

kisz@hpi.de
hpi.de/kisz

1950s–70s

Foundations

How do we optimize decisions over time?

1980s–90s

Learning from Experience

What if the model is unknown?

1990s–2015

Deep RL

How does RL scale to complex states?

2016–2026

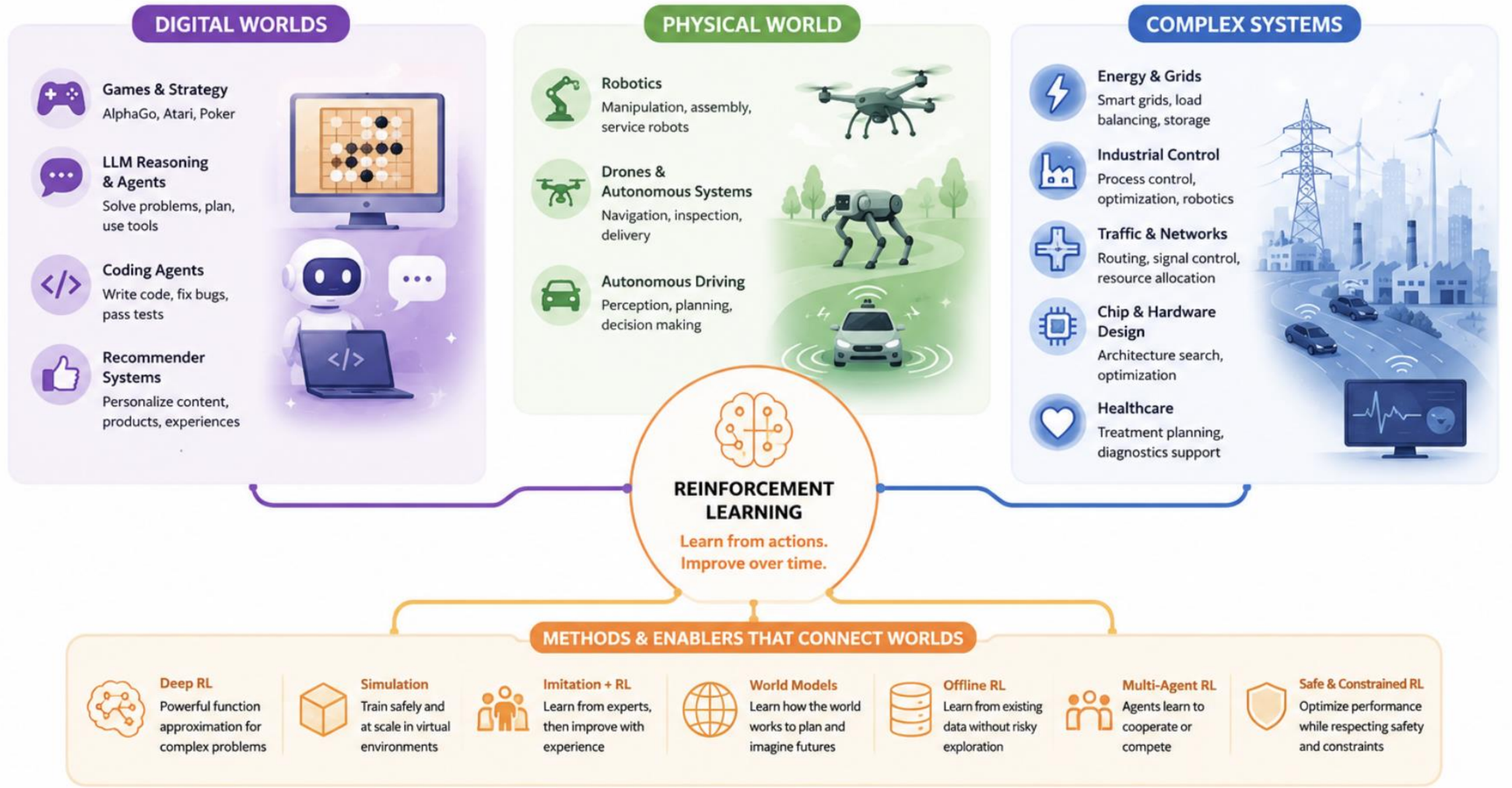
Today & Beyond

Where is RL today?

The Scope of RL Nowadays

Where reinforcement learning is applied today

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz



Autonomous Drone Racing

Can an AI learn to fly faster than a human?

KI Service Zentrum by Hasso-Plattner-Institut
HPI
kisz@hpi.de
hpi.de/kisz

Why RL?

- Continuous decisions at high speed
- Clear goal, but hard to program by hand

What does RL do?

Practise → feedback → better policy

Fast, clean laps are rewarded, crashes are not. Trained in simulation, then flown for real.

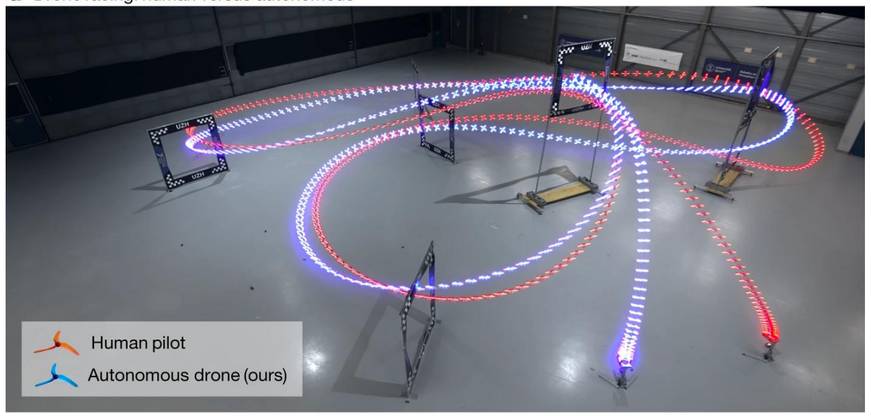
Why does it work?

- Millions of cheap, safe attempts
- Finds strategies humans would not code

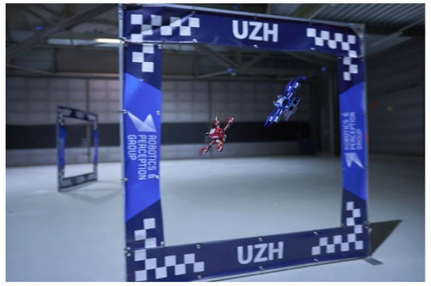
Take away

Swift beat the human world champions in direct races.

a Drone racing: human versus autonomous



b Head-to-head competition



c Human champions

Limitation

Sim-to-real gap limits transfer to reality. New tasks, environments or hardware often require adaptation or retraining.

Sources: Kaufmann et al., Nature (2023), Champion-level drone racing using deep reinforcement learning · Figure: nature.com/articles/s41586-023-06419-4/figures/1

RL for LLMs: Learning From Feedback

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

Can an AI learn which reasoning strategies actually work?

An LLM can produce many answers. Which behaviour should it learn to prefer?

RLHF: Human Feedback

Answers → human preferences → model learns

For goals with no correct answer: helpfulness, tone, preferred behaviour.

RLVR: Verifiable Rewards

Answer → checked automatically → reward

- Math: is the answer correct?
- Coding: do the tests pass?
- Agents: was the task completed?

Take away

RLHF: humans say what they prefer.
RLVR: the task says whether we succeeded.

Limitation

If success cannot be checked, a model may optimise the reward instead of the goal.

Challenges in Applying Reinforcement Learning

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

Why is it difficult to use RL in the real world?

Real experience is expensive

Learning can take thousands or millions of attempts, and a robot cannot crash a million times. Physical practice costs time, money and hardware.

Simulation is not reality

Sensors, friction, weather and people behave differently outside. A policy that works virtually can fail in reality: the sim-to-real gap.

Exploration can be dangerous

Trying out actions is fine in a game, but not in a factory, a car or a clinic. How can an agent explore without causing harm?

Rewards are hard to define

Real goals are more than plus one for success. Drive as fast as possible is not the same as drive safely and efficiently.

The more expensive, unpredictable and safety critical the environment, the harder RL becomes to apply.

When Should You Use Reinforcement Learning?

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

Conclusion: a short checklist before reaching for RL

RL is a good fit when …

✓ **Decisions happen sequentially**
What you do now affects what happens next.

✓ **There is a measurable goal**
Success can be expressed through rewards or feedback.

✓ **The best action is not obvious**
Good behaviour must be discovered through experience.

✓ **Enough experience is available**
Through simulation, existing data or safe interaction.

✓ **Learning from consequences adds value**
A fixed rule, supervised model or classical optimisation is not enough.

Think twice when …

× **Decisions are independent**

× **The correct action is already known**

× **Success cannot be evaluated**

× **Exploration is too expensive or dangerous**

× **A simpler optimisation approach solves the problem**

Interactive: Pen and Paper Exercise

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

Think of a problem where Reinforcement Learning could be useful.

Why does RL make sense, and how would you model it?

Define: Agent · Environment · State · Actions · Reward

Consider: Why RL instead of a simpler approach?

Outlook: What Is Coming in the Next Workshop

From today’s concepts to a running implementation

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

Gymnasium API

Environments, observations, actions and rewards in code.

From problem to RL setup

One example problem, mapped step by step to agent, environment, state, actions and reward.

Algorithms: scope and choice

Which family fits which problem: value based or policy based, model free or model based, on or off policy, plus the theory behind them.

Addressing the challenges

How today’s challenges are handled in practice, from exploration to reward design.

Outlook: Where to go from here

Notebooks, courses and references to keep going

KI Service Zentrum by Hasso-Plattner-Institut

kisz@hpi.de
hpi.de/kisz

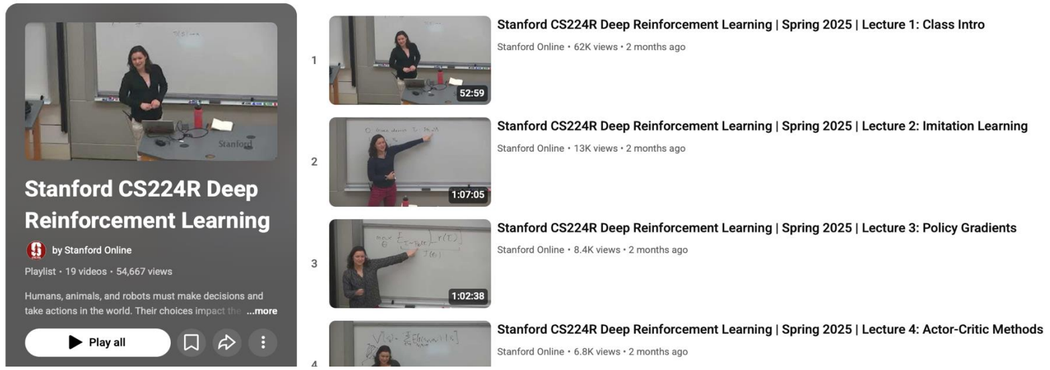
Hands-on notebooks
Examples folder of the workshop repository

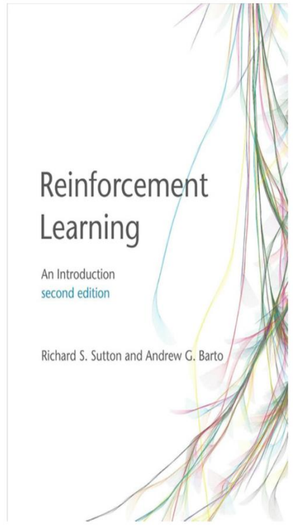
Stanford CS224R
cs224r.stanford.edu — full lecture series

OpenAI Spinning Up
spinningup.openai.com/en/latest

Stable-Baselines3 docs
stablebaselines3.readthedocs.io/en/master

Sutton and Barto book
incompleteideas.net/book/RLbook2020.pdf





KI Service Zentrum by Hasso-Plattner-Institut

HPI

kisz@hpi.de

hpi.de/kisz

Your opinion is relevant!

QR code to feedback form

Gefördert durch:

Bundesministerium für Forschung, Technologie und Raumfahrt